

University of South Florida CDA 3201

Logic Design Lab

Lab Report

Semester: Fall 2022		
Experiment	Number	10
	Date	04/26/2023
Lab	Section	002
	Instructor	Gustavo Olenscki
Report	Due Date	04/28/2023
	Submission Date	04/28/2023
Team	Member #1 name	Diya Jain
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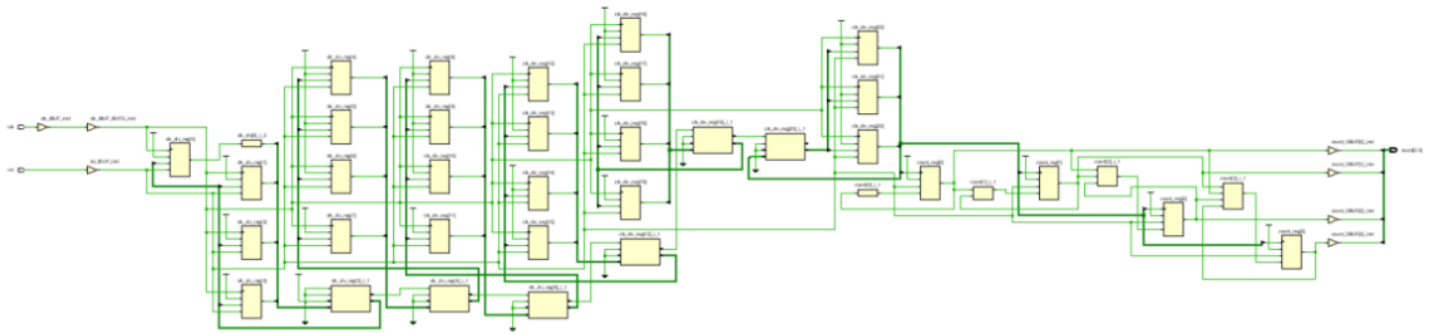
Purpose and objectives:

This lab's goal is to provide students with a quick overview to programmable logic devices, particularly Field Programmable Gate Arrays (FPGAs). A quick Verilog module will be implemented, and the configuration file that results will be mapped to an FPGA development board.

Components used:

- Breadboard
- FPGA
- Micro-USB
- USB cable

Design description:



Observation and data analysis:

1. Cell properties for count[0] i1

Cell Properties

 count[0]_i_1

I0	O=!I0
0	1
1	0

2. Cell Properties for count[1] i1

 count[1]_i_1

I1	I0	O=I0 & !I1 + !I0 & I1
0	0	0
0	1	1
1	0	1
1	1	0

3. Cell properties for count[2] i1

Cell Properties

 count[2]_i_1

I2	I1	I0	O=I0 & I1 & !I2 + !I1 & I2 + !I0 & I2
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

4. Cell Properties for count[3] i1

Cell Properties

 count[3]_i_1

I3	I2	I1	I0	O=I0 & I1 & I2 & !I3 + !I2 & I3 + !I1 & I3 + !I0 & I1 & I3
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

Discussion and conclusion:

The circuit functions as a counter that may be applied to several tasks in the field of digital electronics, such as counting pulses. The team had no trouble putting the circuit together and programming the FPA was simple and uncomplicated.

The FPGA design was extremely user-friendly, which makes it very easy to modify its behavior if we need to change the way it is used. The synchronous counter's simplicity of implementation also makes it clear why it runs quicker than an asynchronous one. Additionally, as it is synchronous and hence regulated by logic gates, there are less chances of mistake.

Answers to the questions:

1. In your own words, explain how an FPGA works. What kind of circuit can it implement?

It is a circuit board that may be configured to carry out tasks according to the requirements and resources offered by a user. In businesses that produce CPUs, it is also used to prototype processors.

2. In the Verilog module, what kind of reset (active high, active low, synchronous, asynchronous) was used? What practical effect did this have on the functionality of the circuit / user interface?

The FPGA was able to reset the counter when pressed down and force it to start counting from 0 all over again since it employed a synchronous (active high) reset.